EE/CprE/SE 4920 STATUS REPORT 1

12/2/25 - 2/5/26

Group Number: sdmay26-40

Project Title: True Force Technologies Arcade Rack

Client: True Force Technologies

Advisor: Matt Post

Team Members/Role:

Jacob Garcia - Frontend Development
Dylan Longlett - Frontend Development
Andy Drafahl - Team Manager, Frontend Development
Fadi Masannat - Frontend Development
Sofi Gutierrez - Backend Development
Parnika Dasgupta - Backend Development
Colin Yuska - Electronics

Weekly Summary

Our group spent this last week getting back into the groove of the project. We spent most of the past two weeks re-familiarizing ourselves with our codebase and planning out how to move forward, especially through communicating with our client, who has come up with several new ideas and potential directions for the project that he would like us to pursue. A few of our team members started to focus on the physical boards that will be running our app.

Past Week Accomplishments

Jacob Garcia

- Helped hook up the new Arduino for the hardwired version.
- Refactored the Git repository.

Parnika Dasgupta

- Looked over the code from the Git Repository
- Tried running the code on my computer for a while, but it kept on not running. I tried to figure out what the problem was and tried fixing it, but noticed that the power puller needed to be connected to the sensor
- Ran the other game on my computer and that worked fine

Dylan Longlett

- Developed power puller game and side game, developed wired connection and Arduino hat
- Working on connecting LEDs.

Fadi Masannat

- I focused on improving our documentation so it matches the updated direction of the project. I updated the current technical documentation to reduce staleness and align it with the split deliverables (strength tester and minigame). I also created a non-technical, high-level conceptual document that explains the system flow and project goals more clearly for non-developers, so it's easier to communicate our progress to the client and faculty.

Sofi Gutierrez

- I looked over the code from the folder PowerPuller3000, where the main game logic and style are. I took the style from the main game to create a mock-up of the minigame in Python. It resembles the game Flappy Bird in style of the main game

Colin Yuska

- Researched arcade hobby sites to find good resources when I start breaking into the arcade cabinet.

Andy Drafahl

- Developed Gantt Charts for all three teams (electronics, minigame, and strength tester) up until the end of the semester.
- Updated animation details and sent them to the client for final approval.
- Pulled down the current strength tester code to run locally as a base for the minigame, but the connection to the load sensors needs to be bypassed first.

Individual Contributions

Name	Individual Contributions	Hours This Week	Hours Cumulative
Jacob Garcia	Helped hook up the new Arduino for the hardwired version. Git management.	2	2
Parnika Dasgupta	Played around with the existing code and tried to understand it. Made some plans about how to proceed with making the mini game	1.5	1.5
Dylan Longlett	Developed a power puller game and a side game, developed a wired connection, and an Arduino hat	8.67	8.67
Sofi Gutierrez	Created a mock minigame in the style of our strength game	3	3
Colin Yuska	Researched arcade hobby sites and blogs to find repair guidelines and similar projects	2.75	2.75
Fadi Masannat	Updated our current technical documentation and created a non-technical high-level conceptual documentation.	4	4
Andy Drafahl	Developed Gantt Charts for all three teams and sent the client final animation details to approve, pulled code to run locally.	2.03	2.03

Plans for the Upcoming Week

Fadi Masannat

- Create and generate some supplemental visuals and store them. Integrate the Strength tester with the current documentation after it gets added to the project repository. I also want to deeply understand the strength tester codebase, as I have been assigned the custom mode for all of last semester.

Jacob Garcia

- Refine the strength tester.

Dylan Longlett

- Continue work on games and wired connection stuff

Colin Yuska

- Begin research on similar arcade cabinets for repairs
- Find comparable projects to begin ideas for arcade cabinet additions

Sofi Gutierrez

- Will continue working on the mini-game logic

Parnika Dasgupta

- Will get together with Sofi and figure out how to work on the mini game separately as well as as a team

Andy Drafahl

- Will work with Sofi and Parnika to refine the minigame and adjust difficulty settings and game mechanics.

Summary of Weekly Advisor Meeting

Pulled from our meeting notes:

- Ben has a couple of arcade cabinets and could bring them in. (All from the snow)
- Matt recommends looking at Chinese sites for these displays. Displays with no case.
- The car dashboard will be removed, and cabinets will be put there.
- Possibly could use tools here at SICTR, just there are maybe 10 hours of training sessions you have to do to get full access.
- Notes on document from Matt:
  - Said he appreciates that we were given a really open-ended project and that we completed a lot of deliverables from it.
  - “Just don’t fall behind. Keep doing what you’re doing.”
- Julian Baldi is coming to Ames to test out the rack. Ben is flying him out.
- He is going to move the bar a little bit towards the cabinet so that users can lift it facing the screen or away from it.

- We could actually do retro sounds if we get IC's (although Andy is treating those as a stretch goal.